LINUX FUNDAMENTALS – WEEK 1 LECTURE NOTES

Introduction to Linux

Linux is an open-source, Unix-like operating system developed by Linus Torvalds. It is widely used in servers, cloud computing, embedded systems, cybersecurity, and software development.

Features of Linux

- Open Source
- Multiuser
- Multitasking
- Secure
- Portable
- Stable
- Command Line Support
- GUI Support

Learning Objectives

- Understand Linux architecture.
- Navigate the Linux filesystem.
- Use command-line tools effectively.
- Manage files and directories.
- Configure Linux systems.
- Install and manage software packages.
-

Linux Philosophy and Concepts

Core Concepts

Open Source Software

Definition: Source code is publicly available and can be modified or redistributed.

The source code is publicly available and can be modified or distributed by anyone.

Linux Architecture

Hardware → Kernel → Shell → User Applications

Definition: Core of Linux that manages hardware, memory, processes and files.

Definition: Interface between the user and the Linux kernel.

Kernel

Definition: Core of Linux that manages hardware, memory, processes and files.

The kernel is the core component of Linux responsible for:

- Process Management
- Memory Management
- Device Management
- File System Management

Shell

Definition: Interface between the user and the Linux kernel.

The shell acts as an interface between the user and the kernel.

Examples:

- Bash
- Zsh
- Fish

Linux Principles

- Everything is a File
- Small Modular Tools
- Command-Line Efficiency
- User Permissions and Security
- Flexibility and Portability

Linux Basics and System Startup

Linux Boot Process

1. Power On
2. BIOS/UEFI Initialization
3. Bootloader (GRUB)

Definition: Firmware that initializes hardware during startup.

Definition: Modern replacement for BIOS.

4. Kernel Loading

Definition: Program that loads the Linux kernel.

5. System Initialization
6. Service Startup
7. User Login

Important Terms

BIOS

Basic Input Output System used to initialize hardware during startup.

UEFI

Definition: Modern replacement for BIOS.

Modern replacement for BIOS with enhanced features.

Definition: Firmware that initializes hardware during startup.

Bootloader

Definition: Program that loads the Linux kernel.

Loads the Linux kernel into memory.

Definition: Core of Linux that manages hardware, memory, processes and files.

Example: GRUB (Grand Unified Bootloader)

Daemon

Definition: Background process running continuously.

Background process running continuously in Linux.

Examples:

- sshd
- cron
- systemd

Startup Commands

Check system information:

```
uname -a
```

Check kernel version:

```
uname -r
```

Shutdown system:

```
sudo shutdown now
```

Restart system:

```
sudo reboot
```

Graphical Interface

Desktop Environment

Definition: GUI providing windows, icons and menus.

A Desktop Environment provides graphical interaction with the system.

Popular Desktop Environments:

- GNOME

- KDE Plasma
- XFCE
- Cinnamon

Features

- Desktop
- File Manager
- Panels
- Workspaces
- Application Launcher
- Settings Manager

Desktop Operations

- File Browsing
- Window Management
- Display Configuration
- Theme Customization
- Workspace Management

System Configuration Using GUI

User Management

Add User:

```
sudo adduser username
```

Delete User:

```
sudo deluser username
```

Network Configuration

View IP Address:

```
ip a
```

Check Hostname:

```
hostname
```

Change Hostname:

```
sudo hostnamectl set-hostname newname
```

System Monitoring

View Running Processes:

```
top
```

View Memory Usage:

```
free -h
```

View Disk Usage:

```
df -h
```

Common Linux Applications

Web Browsers

- Firefox
- Chromium

Office Applications

- LibreOffice

Media Players

- VLC Media Player

Text Editors

- Nano
- Vim
- Gedit

Development Tools

- VS Code
- Eclipse
- IntelliJ IDEA

Compression Utilities

- zip
- unzip
- tar

Command Line Operations

Terminal Basics

Open Terminal:

```
Ctrl + Alt + T
```

Current User:

```
whoami
```

Current Directory:

```
pwd
```

Display Date:

```
date
```

Display Calendar:

```
cal
```

Clear Terminal:

```
clear
```

Directory Navigation

List Files and Directories:

```
ls
```

Detailed Listing:

```
ls -l
```

Show Hidden Files:

```
ls -la
```

Change Directory:

```
cd directory_name
```

Move to Home Directory:

```
cd ~
```

Move Back One Directory:

```
cd ..
```

Move to Root Directory:

```
cd /
```

Print Working Directory:

```
pwd
```

File and Directory Management

Create File

```
touch file.txt
```

Create Directory

```
mkdir myfolder
```

Remove Empty Directory

```
rmdir myfolder
```

Remove File

```
rm file.txt
```

Remove Directory Recursively

```
rm -r myfolder
```

Copy File

```
cp source.txt destination.txt
```

Move/Rename File

```
mv oldname.txt newname.txt
```

Viewing File Contents

Display Entire File:

```
cat file.txt
```

View File Page by Page:

```
less file.txt
```

View First 10 Lines:

```
head file.txt
```

View Last 10 Lines:

```
tail file.txt
```

Hard Links and Symbolic Links

Definition: Another filename pointing to the same inode.

Definition: Shortcut pointing to another file or directory.

Hard Link

```
ln file1 hardlink
```

Symbolic Link

```
ln -s file1 symlink
```

List Inode Numbers:

```
ls -li
```

Searching for Files

Locate Command

Definition: Quickly searches files using a database.

Search File:

```
locate filename
```

Update Database:

```
sudo updatedb
```

Find Command

Definition: Searches files/directories using conditions.

Find File by Name:

```
find /home -name file.txt
```

Find Directories:

```
find /home -type d
```

Find Large Files:

```
find / -size +10M
```

Find Files Modified in Last Day:

```
find /home -mtime -1
```

Wildcards

Definition: Special characters used for pattern matching.

All Text Files:

```
ls *.txt
```

Files Starting with “a”:

```
ls a*
```

Single Character Match:

```
ls file?.txt
```

Input and Output Redirection

Redirect Output:

```
ls > output.txt
```

Append Output:

```
ls >> output.txt
```

Redirect Input:

```
cat < input.txt
```

Redirect Errors:

```
command 2> error.txt
```

Pipes

Definition: Sends output of one command as input to another.

A pipe passes output from one command as input to another.

Example:

```
ls -l | less
```

Count Files:

```
ls | wc -l
```

Search Text:

```
cat file.txt | grep Linux
```

User Privileges and Sudo

Execute Command as Administrator:

```
sudo command
```

Switch User:

```
su username
```

Switch to Root:

```
sudo -i
```

Package Management

APT Package Manager (Ubuntu/Debian)

Definition: Tool to install, update and remove packages.

Update Package List:

```
sudo apt update
```

Upgrade Installed Packages:

```
sudo apt upgrade
```

Install Package:

```
sudo apt install package_name
```

Remove Package:

```
sudo apt remove package_name
```

Search Package:

```
apt search package_name
```

List Installed Packages:

```
apt list --installed
```

Lab Exercises Completed

Lab 7.1

Killing the Graphical User Interface

Lab 7.2

Locating Applications

Lab 7.3

Creating, Moving and Removing Files

Lab 7.4

Finding Directories and Creating Symbolic Links

Definition: Shortcut pointing to another file or directory.

Lab 7.5

Installing and Removing Software Packages

Key Takeaways

- Learned Linux architecture and operating principles.
- Understood Linux boot process and startup sequence.
- Explored graphical desktop environments.
- Configured Linux systems using GUI tools.

  Definition: GUI providing windows, icons and menus.
- Practiced command-line operations.
- Managed files and directories efficiently.
- Used links, wildcards, pipes, and redirection.
- Installed and removed software packages.

  Definition: Special characters used for pattern matching.

  Definition: Sends output of one command as input to another.
- Gained hands-on experience through laboratory exercises.
- Created a GitHub repository to document internship activities and learning progress.